

The Metric Is Not the Construct

Construct validity at every layer of a lending audit

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1 THE QUESTION

When a fairness audit prints a number, say “disparate impact 0.94,” does that number measure what its name says? We built a complete fairness audit of consumer credit and mortgage models (1,000 and 10,978 real applications, every applicant scored out-of-fold), and then audited the audit.

2 THE RESULT

The same model, judged on the same applicants, passes or fails depending on settings that appear nowhere in the output. Nine measurements in this one audit computed something other than the construct they were named for, and every one produced a clean, reassuring result. The four sharpest:

(1) A library constructor default (favorable_label=1.0, AIF360 0.6) decides which outcome of a loan is *good*. On a default-coded target it reports the disadvantaged group as favoured: DI = 1.2072 where the oriented value is 0.9862.

(2) The age attribute decides which age question is asked. Banded at 40+ (a threshold imported from employment practice) the model reads 0.9791: clean. On the 62+ boundary the mortgage data ship a dedicated flag for, *the same model over the same people* reads **0.8897**, the largest age disparity present. The public age bands straddle 62, so no banded attribute can ever see it.

(3) The reporting policy decides what small groups can do to the verdict. At the audit size we first used ($m=2,196$ of 10,978), 2,000 re-sampled audits show: point estimates falsely clear a truly-violating subgroup in 83.8% of draws; reporting floors never false-clear but name the wrong worst-treated group in 100% (they never report it); honest intervals must abstain in 90.3%. *Every* policy that shows small cells is usually wrong; every policy that hides them is silently blind.

(4) Our own checking instruments did it too: a refusal detector defining refusal as *under 40 characters* flagged six correct terse answers and helped block a provider, and most of a cross-model “prompt sensitivity” spread (14.7 vs. 0.0 points) came from our scorer penalising instructed brevity; excluding that cycle, 6.7.

3 WHAT IT MEANS

A fairness verdict is therefore not a property of the model alone; it is a property of the model plus a configuration (label orientation, attribute banding, reporting policy, reference group, detector thresholds) that is set silently and appears nowhere in the output. Two consequences. **A fairness number you are handed is uninterpretable without its configuration:** the same data honestly support “worst cell 0.67,” “worst cell 0.90,” or “worst cell unknownable.” And **the errors point one way:** every wrong verdict we measured erred toward false clearance, the reassuring direction nobody rechecks. An audit should ship its configuration the way results ship error bars; the section below is that record.

4 WHAT AN AUDIT MUST RECORD

The favorable outcome, stated explicitly. The interval, n , and reference group beside every ratio, plus every suppressed cell. The definition’s source beside every threshold. The reporting policy by name. Population construction (we dropped 45.1% of the raw file before any metric ran). And an adversarial test for every mechanical gate, starting where the correct answer is unusual. None of this is hard; all of it is currently optional.

5 THE PROOF

Table 1: Every claim regenerates offline from the released repository.

Claim	Where to check it
(1) default 1.2072 vs. oriented 0.9862	pinned repro: scripts/aif360_default_repro.py
(2) banded 0.9791 vs. 62+ 0.8897 (same 10,978 applicants)	hmda/fairness/fairness_summary.json
(3) false clearance 83.8%; floors wrong 100%; abstention 90.3%	scripts/subsample_study.py
(4) spread 14.7 → 6.7 excl. the terse cycle; 6 flags, 0 refusals	scripts/recheck_prompt_sensitivity.py
6/12 cells band-indeterminate; 0.6734 [0.45, 0.90] at $n=31$	scripts/interval_estimates.py
45.1% attrition before any metric	scripts/hmda_attrition.py

86 tests, CI green; the deterministic layer regenerates with no API key.